

ME597: Machine Learning (ML)

Tutorial 6

Week 5

School of Mechanical Engineering
Purdue University

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Machine Learning Algorithms

Machine Learning

Supervised Learning

- Regression (ML 5)
- **ANN (Artificial Neural Network)** → **Deep Learning**
- ARIMA, Exponential Smoothing
→ Mainly for prediction (also classification)
- KNN (K-Nearest Neighbor)
- SVM (Support Vector Machine)
- Decision Tree
- Random Forest
→ Mainly for classification (also prediction)

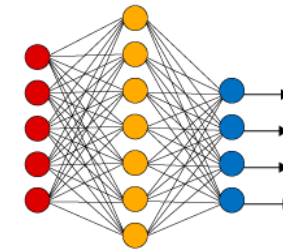
Unsupervised Learning

- K-means Clustering
- Density-based Clustering
- Fuzzy Clustering
→ Mainly for clustering (classification without labels)
- Association Rule Analysis
- Network Analysis
→ Mainly for data structure/pattern recognition
- **PCA(Principal Component Analysis)**
- **t-SNE(Stochastic Neighbor Embedding)**] (ML 4)
→ Mainly for dimensionality reduction (visualization)

Deep Learning

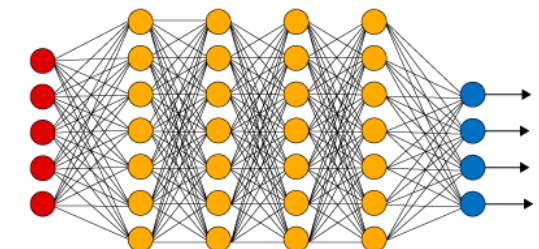
- **CNN (Convolutional Neural Network)** (ML 6)
- RNN (Recurrent Neural Network)
- AE (Autoencoder)
- GAN (Generative Adversarial Network)
- Transformer
- ...
- Generative Pre-trained Transformer (GPT)

Simple Neural Network



● Input Layer

Deep Learning Neural Network



● Hidden Layer

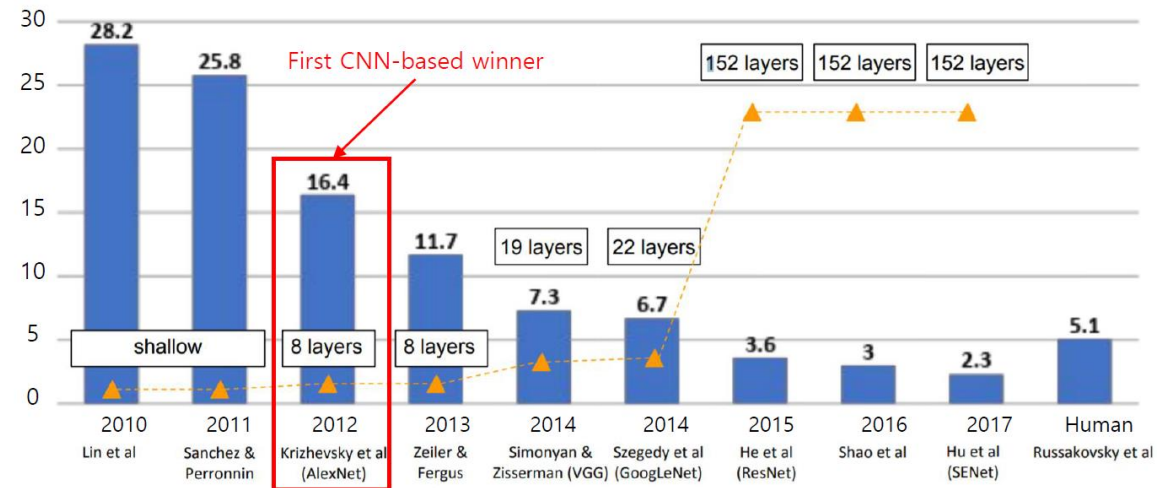
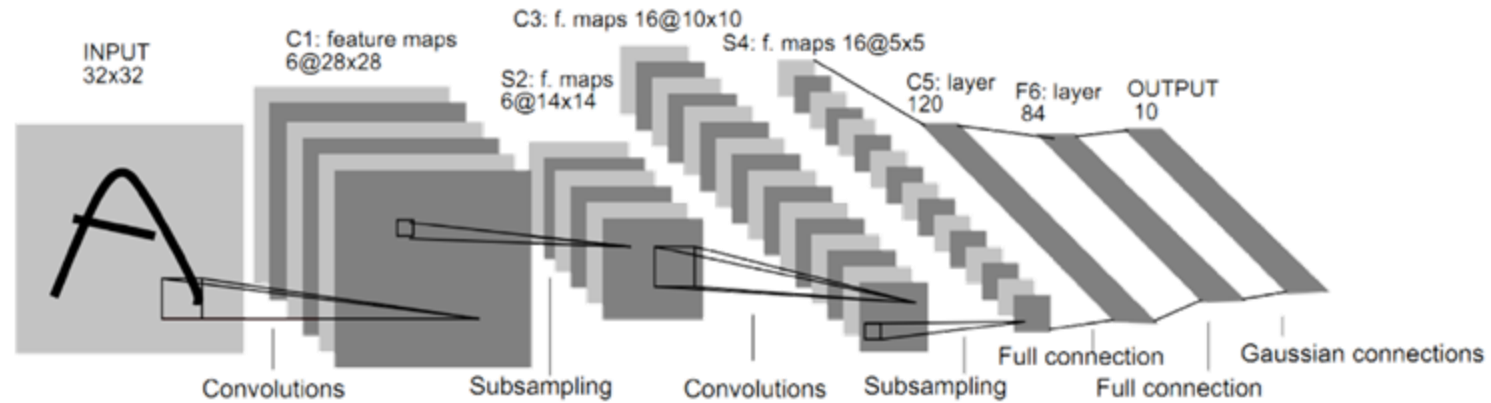
● Output Layer

Convolutional Neural Network (CNN)

* **LeNet (=CNN)**, Yann LeCun, et al., *Proceedings of IEEE* (1998), "Gradient-based learning applied to document recognition"



Yann LeCun
(AT&T, USA)

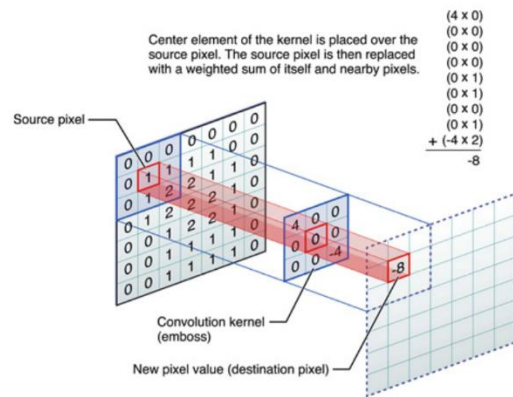
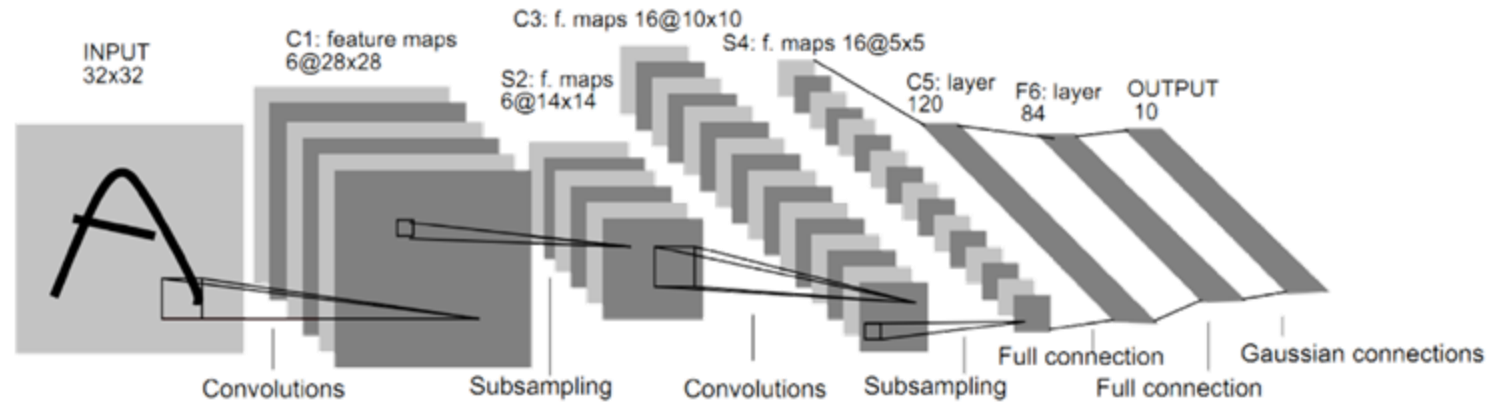


Convolutional Neural Network (CNN)

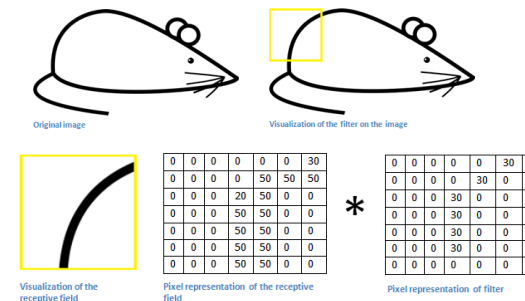
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Yann LeCun
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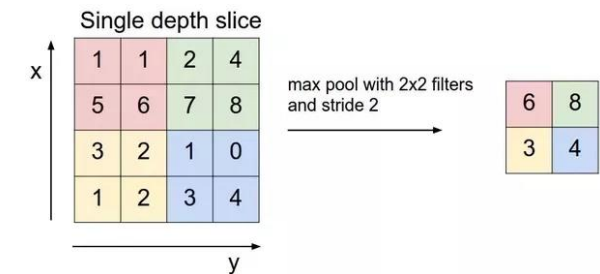


Convolution



Multiplication and Summation = $(50 \times 30) + (50 \times 30) + (50 \times 30) + (20 \times 30) + (50 \times 30) = 6600$ (A large number!)

Pooling (Subsampling)



Convolution

0	1	1
1	1	0
0	1	0

<Image>

*

1	1	0
1	0	1
0	1	0

<Kernel>

=

$$\begin{pmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{pmatrix} * \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix}$$

$$= (I_{11} \ I_{12} \ I_{13} \ I_{21} \ I_{22} \ I_{23} \ I_{31} \ I_{32} \ I_{33}) \cdot (K_{11} K_{12} K_{13} K_{21} K_{22} K_{23} K_{31} K_{32} K_{33})$$

$$= \sum_{i=1}^3 \sum_{j=1}^3 I_{ij} K_{ij}$$

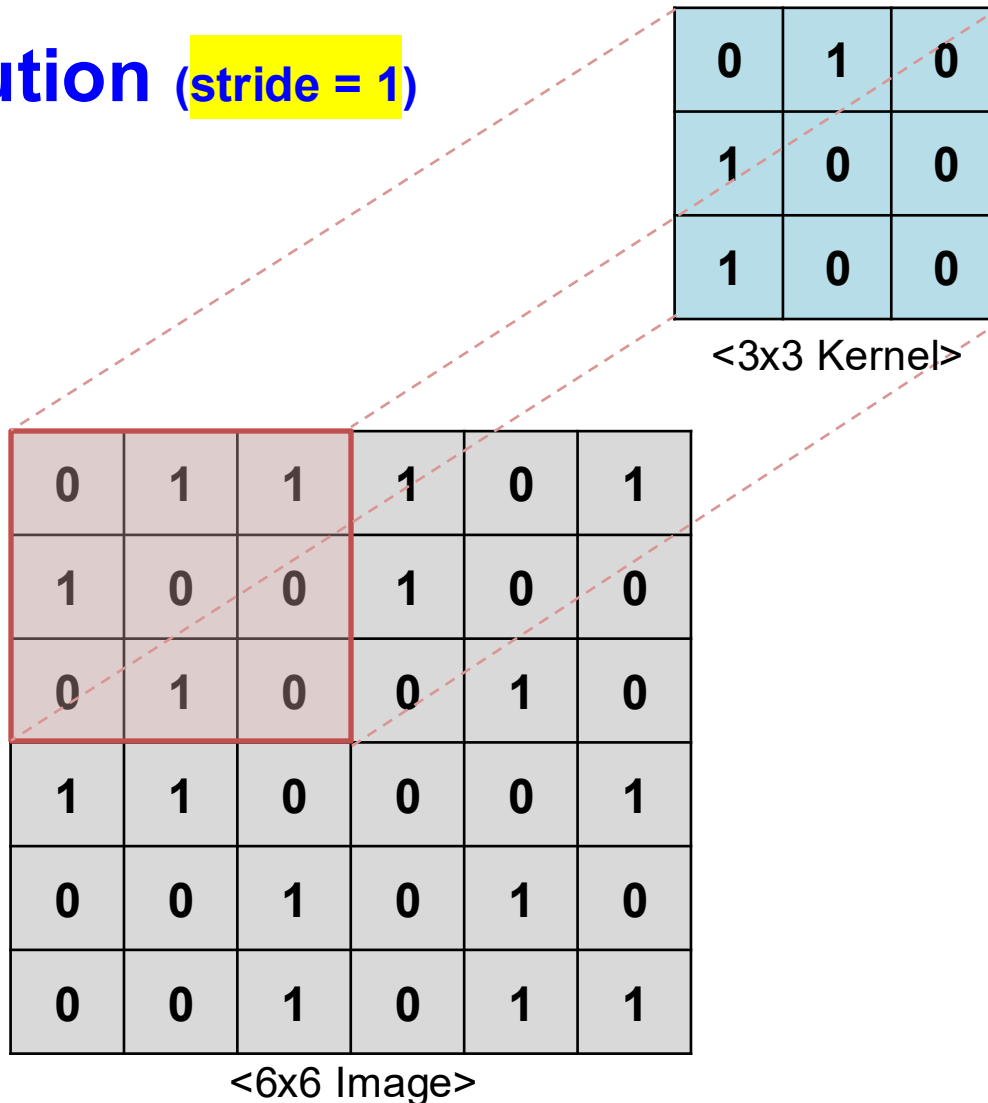
$$= 3$$

Generalization

$$\rightarrow I * K = \sum I_{ij} K_{ij}$$

Convolutional Neural Network (CNN)

Convolution (stride = 1)



$$I * K = \sum I_{ij} K_{ij}$$

Convolutional Neural Network (CNN)

Convolution (stride = 1)

0	1	0
1	0	0
1	0	0

<3x3 Kernel>

$$I * K = \sum I_{ij} K_{ij}$$

0	1	1	1	0	1
1	0	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1
0	0	1	0	1	0
0	0	1	0	1	1

<6x6 Image>

	2	2			

Convolutional Neural Network (CNN)

Convolution (stride = 1)

0	1	0
1	0	0
1	0	0

<3x3 Kernel>

$$I * K = \sum I_{ij} K_{ij}$$

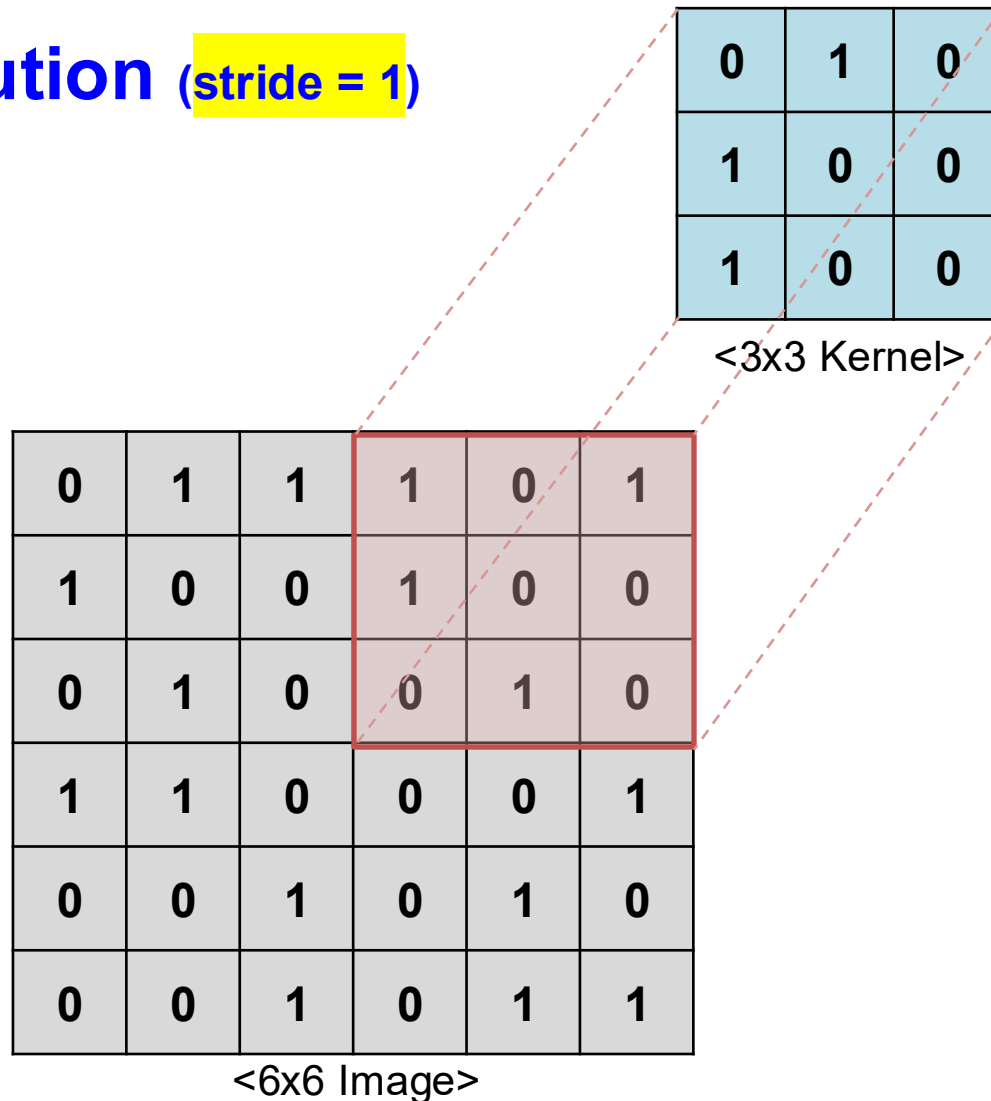
0	1	1	1	0	1
1	0	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1
0	0	1	0	1	0
0	0	1	0	1	1

<6x6 Image>

	2	2	1		

Convolutional Neural Network (CNN)

Convolution (stride = 1)



$$I * K = \sum I_{ij} K_{ij}$$

	2	2	1	1	

Convolutional Neural Network (CNN)

Convolution (stride = 1)

0	1	0
1	0	0
1	0	0

<3x3 Kernel>

$$I * K = \sum I_{ij} K_{ij}$$

0	1	1	1	0	1
1	0	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1
0	0	1	0	1	0
0	0	1	0	1	1

<6x6 Image>

	2	2	1	1	
	1				

Convolutional Neural Network (CNN)

Convolution (stride = 1)

0	1	0
1	0	0
1	0	0

<3x3 Kernel>

$$I * K = \sum I_{ij} K_{ij}$$

0	1	1	1	0	1
1	0	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1
0	0	1	0	1	0
0	0	1	0	1	1

<6x6 Image>

	2	2	1	1	
	1	2			

Convolutional Neural Network (CNN)

Convolution (stride = 1)

0	1	0
1	0	0
1	0	0

<3x3 Kernel>

0	1	1	1	0	1
1	0	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1
0	0	1	0	1	0
0	0	1	0	1	1

<6x6 Image>

$$I * K = \sum I_{ij} K_{ij}$$

m x m Image

n x n Kernel

→ **Feature size : (m-n)/stride + 1**

	2	2	1	1	
	1	2	1	0	
	2	1	1	1	
	1	0	2	0	

Convolutional Neural Network (CNN)

Pooling

0	1	1	1	0	1
1	0	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1
0	0	1	0	1	0
0	0	1	0	1	1

<6x6 Image>

2	2	1	1
1	2	1	0
2	1	1	1
1	0	2	0

<4x4 Feature>

Pool with **2x2 filters**
and **stride 2**

Feature size : $(4-2)/2 + 1 = 2$
→ **2x2 Feature**

Max Pooling

2	1
2	2

<2x2 Feature>

Average Pooling

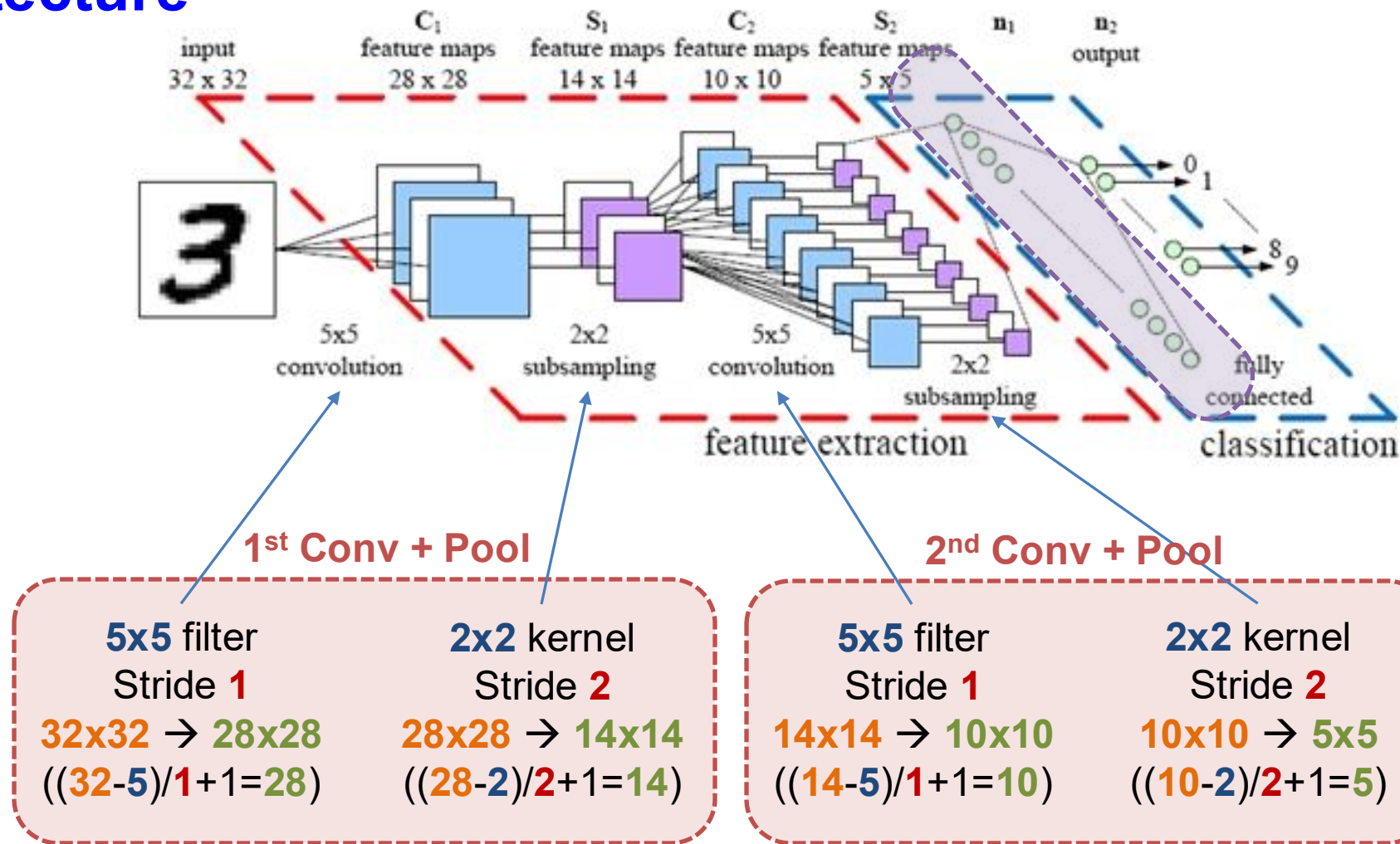
1.75	0.75
1	1

<2x2 Feature>

Convolutional Neural Network (CNN)

CNN architecture

Flatten → Dense layers (=ANN)



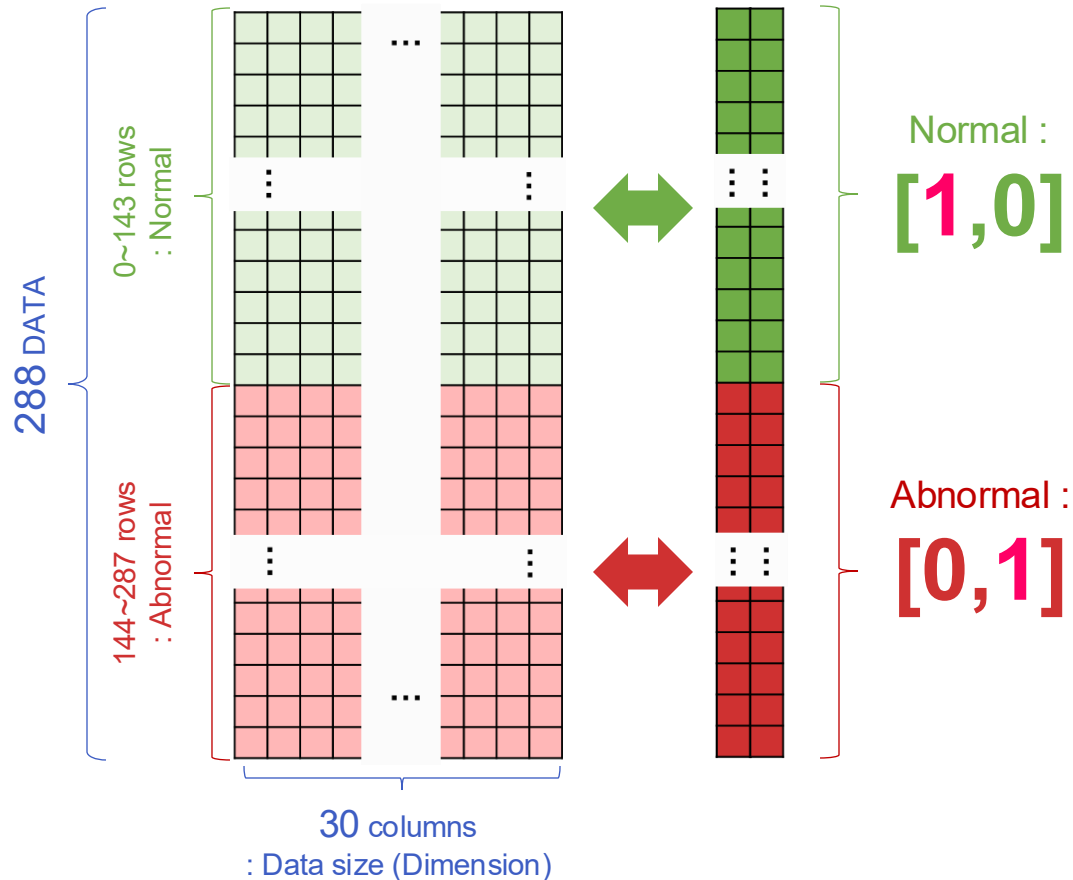
Data Labeling for ANN (One-Hot Encoding)

Training Dataset (80%)

288 rows X **30** columns

Training Label

288 rows X **2** columns



Test Dataset (20%)

72 rows X **30** columns

Test Label

72 rows X **2** columns

